CV4 — Combined MLOps Architecture

Data pipeline → training pipeline → deployment (on-premise + cloud), closed by feedback & retraining loops

DATA PIPELINE

NPEC / HADES camera
Captures plant images in petri dishes

Manual annotation
Label root structures → TIF masks

Raw images + masks
PNG images + TIF root masks

1 · Match images ↔ masks
By filename · skip unmatched

2 · Crop petri dish
Detect bright region → crop image + mask

3 · Pad to patch size
Pad to nearest multiple of patch size

4 · Cut into patches
Overlapping patches for model input

5 · Train / validation split
Split into training and validation sets

Azure Blob Storage
Versioned data assets (v1, v2, v3…)

User feedback
Corrected masks from researchers

corrected / new data

versioned data

MODEL TRAINING PIPELINE

Training trigger
Weekly · new data · feedback threshold

1 · Load versioned data
Train/val/test patches from Azure

2 · Preprocess + augment
Normalize · flip/rotate/zoom · balance

3 · Train Residual U-Net + Attention
TensorFlow · Adam · hybrid loss · F1

Hyperparameter tuning
Azure ML sweep · best run auto-registers

4 · Evaluate on validation set
F1 · loss · precision · recall

5 · Log metrics to tracker
Azure ML / MLflow · curves · examples

Is new model better?
Compare new F1 vs deployed F1

NO → **Keep current model**
Discard new · current stays

YES

Register new model version
Save v1, v2… in Azure ML registry

Deploy to production
Update Azure ML endpoint

Monitor model performance
F1 over time · drift · collect feedback

drift / new data → retrain

DEPLOYMENT / CI-CD PIPELINE

Push code to GitHub
Feature branch → PR → code review

Ruff lint check
Style · unused imports · errors

Run pytest
Preprocessing · inference · API

Build Docker containers
Backend (FastAPI) · Frontend · Database

All checks passed?

NO → **Rejected**
Fix code and push again

YES

Human approval
Reviewer approves before production deploy

registered model → served as endpoint

On-premise — Portainer
Backend + Frontend + DB
BUas server

Cloud — Microsoft Azure
Container Apps · ML Endpoint
Database · Blob Storage

Monitor both deployments
Latency · errors · accuracy · feedback · rollback

prod metrics / drift → retrain

user feedback

Source / Trigger · Processing · Data / Tracking · Cloud (Azure) · CI/CD step · Decision · Reject · Feedback loop

— — dashed = feedback / retrain loop ——— solid = forward flow / hand-off